

Studying the effect of seasonal changes on the infrared radiation properties of the ground target using multi-physics simulation

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ABSTRACT

The infrared radiation properties of ground targets are influenced by the surrounding environment, solar radiation and climate. Typically, the target and its environment exhibit different temperature distributions due to different thermodynamic characteristics. Coupled multi-physics field simulation method are used, and the infrared radiation characteristics of ground target is investigated under different seasons based on thermodynamics and computational fluid dynamics. The temperature contour plots are observed to find that the temperature distributions of the ground target and its background are closely related to the seasons. The temperature variation curves of different monitoring points are compared to obtain the temperature of the ground target with respect to ambient temperature, structures, materials and solar altitude angle.

Introduction

Any object with a temperature above absolute zero emits radiant energy at all times, and most of them are in the infrared band with wavelengths in the range of 0.78–1000 μm , which are located outside the red light [1]. Since infrared detection has passive tracking characteristics and can be used to detect information such as target position, shape and state, the thermodynamic properties of the target are of great importance for research [2].

Today infrared radiation has a wide range of applications in various fields, in civil applications such as atmospheric monitoring, biomedicine and urban planning. The effects of infrared radiation on water separation in the processing of agricultural products were investigated by Zeng et al [3]. Tatartchenko et al. used infrared radiation to detect the results of water crystallization, condensation and deposition in atmospheric phenomena [4]. Botchway et al. studied the effects of near-infrared femtosecond lasers on cellular DNA damage and repair [5]. Blunk outlined the role of infrared radiation in the flow of participating media, including quantitative measurements, visualization of thermographic flow and evaluation of CFD simulation results [6]. Martin et al. used infrared thermography to study urban heat fluxes and used multi-scale thermal images to build a better model of urban energy [7].

Military aspects such as infrared stealth, infrared detection and thermal imaging have also become popular research in various countries. Mahulikar et al. studied the infrared properties of aerospace vehicles and illustrated the relationship between infrared signature levels and target sensitivity [8]. Li et al. found that rotor flow has a significant effect on the temperature distributions and infrared radiation characteristics of helicopters, and used ray-tracing methods to study the infrared radiation from the solid walls of helicopters [9]. Li et al. developed accurate models for characterizing the infrared radiation of rocket plumes [10]. Lang et al. designed convolutional neural networks, which enhanced filter targets and provided background suppression for infrared images [11]. Swiderski used infrared thermography to achieve nondestructive detection of military buildings and light armor [12].

Infrared radiation is of great significance and outstanding research value in various industries. Currently, the research methods for infrared radiation characteristics are mainly divided into measurement-based experimental methods and theory-based computational methods. Although the experimental method is intuitive and accurate, it often consumes a lot of human and material costs, so numerical simulation methods based on thermodynamics, optics and computational fluid dynamics have been widely used [13–15]. A comprehensive, visualized infrared radiation signature and its comparative characteristics can be

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obtained through finite element simulation analysis. Mahdi et al. used ANSYS FLUENT to study the temperature distribution at a wind turbine airport in a high temperature environment [16]. Yang et al. used coupled CFD simulations and infrared radiation intensity simulations to investigate the effect of helicopter forward flight speed on plume and infrared radiation [17].

Scholars have conducted comprehensive studies and analyses in the field of infrared radiation, but some of these studies have focused on the target itself and ignored the surrounding environment, while others have studied the static infrared radiation properties of the target without conducting time-domain studies. In fact, ground targets with different background temperatures around them emit different radiation intensities and distribution patterns of thermal radiation, and exhibit different thermodynamic and infrared radiation properties. In addition, the temperature and infrared radiation characteristics of ground targets change with the environment and time. Therefore, it is necessary to carry out the study of infrared radiation characteristics of ground targets with their environment.

Based on thermodynamics and computational fluid dynamics, this paper used a coupled multi-physics field approach to study the thermodynamic properties of the van in a background environment with objects such as hills, trees and buildings. Three days with typical spring, summer and winter characteristics were selected for dynamic simulation calculation respectively. The effect of seasonal changes on the thermodynamic characteristics of the target was carefully studied by observing the change pattern of temperature distributions of the ground target and the surrounding environment and comparing the temperature curves of the monitoring points. This study may provide useful information and help for future studies on the infrared radiation characteristics of targets and backgrounds.

Physical model

Model parameters

In the current study, a van model was assumed to be the ground target, and the detailed parameters of the van model are shown in the Table 1. The length of the model is 5995 mm, the width is 2000 mm, and the height is 2715 mm. The model is equipped with three different shapes of glass, including rectangular glass in the front with a length of 1880 mm and a width of 784 mm, trapezoidal glass on the side and square glass. The tires of the model have a diameter of 900 mm and a thickness of 200 mm. The van model is shown in Fig. 1.

The effect of the internal parts of the van on the thermodynamic properties was not considered in this study, so the interior of the model is a cavity filled with air. The body of the van model is a carbon steel material with a density of 7768.98 kg/m³, a constant pressure heat capacity of 460.967 J/(kg·K), and a thermal conductivity of 52.019 W/(m·K). The windows are silica glass with a density of 1450 kg/m³, a constant pressure heat capacity of 817 J/(kg·K), and a thermal conductivity that is usually not constant, taking its peak at 1.6528 W/(m·K).

Table 1

Parameters of the simulation model.

Specific	Parameter	Value (mm)
Body	Length	5995
	Width	2000
	Height	2715
Glass	Length of front glass	1880
	Width of front glass	784
	Length of the upper side of the trapezoidal glass	450
	Length of the lower side of the trapezoidal glass	800
	Height of the trapezoidal glass	550
	Length of square glass	700
	Thickness of glass	70
Tires	Diameter of tires	900
	Thickness of tires	200

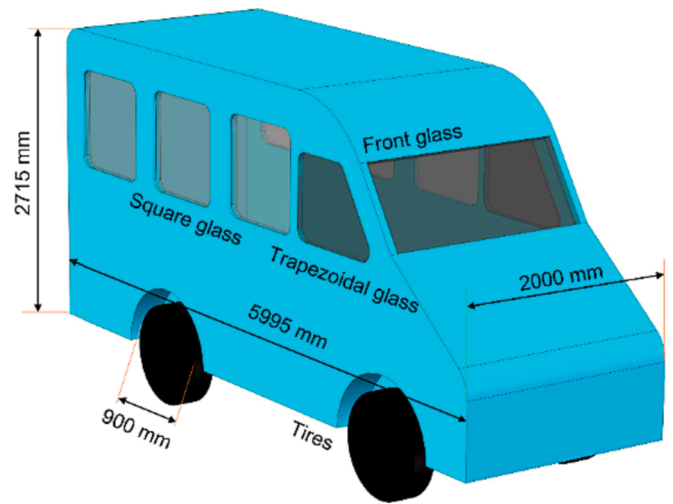


Fig. 1. Van(ground target) model for simulation.

Tires are rubber materials with a density of 1000 kg/m³, a constant pressure heat capacity of 1700 J/(kg·K), and a thermal conductivity of 0.18 W/(m·K).

The thermodynamic and infrared radiation properties of a ground target are not isolated to itself but also related to the environment in which the target is located, including solar irradiation, atmospheric absorptivity, ground reflectivity, radiation from surrounding objects, and air flow. The ground target is placed in an environment consisting of concrete ground, sandy hills, buildings and trees, as shown in Fig. 2. The entire site is 200 m long and 250 m wide with a concrete floor thickness of 5 m. The sun rises from the side of the hill, which is also the left side of the van, and sets from the right side of the van. In addition, six temperature sensors were set up on different surfaces of the van to record temperature changes, with point CH1 at the center of the top of the van, points CH2 and CH3 on the right and left sides of the body, points CH4 and CH5 on the square windows on the right and left sides, respectively, and point CH6 at the center of the rear side of the van.

Model assumptions

To simplify the analysis, the following assumptions are used: 1) The weather is always sunny and the ambient temperature and ambient wind speed are dynamic, but the clear sky index and atmospheric transmittance remain constant. 2) The natural convection of the air inside the target due to temperature change gradients and gravity is ignored. 3) The latent heat exchange due to air humidity and surface moisture content, moisture evaporation and condensation is ignored.

Numerical simulation methods

Control equations

The coupled multi-physics field simulation of ground targets involves solid and fluid heat transfer, radiative heat transfer, turbulent flow, and non-isothermal flow problems. Among them, heat transfer occurs inside solid structures such as van, buildings, trees, and the ground, convective heat transfer occurs at the target surface, mainly forced external convection influenced by the ambient wind field, and thermal radiation comes from solar radiation, atmospheric scattering, and radiation from the ground and other objects. In this study, a split-coupling solver is used to split the solution process into multiple sub-steps, each using a damped version of the Newtonian method. The calculations were performed using COMSOL Multiphysics software, and the Reynolds-averaged Navier-Stokes equations were used in the computational fluid dynamics analysis to obtain the properties of the flow and temperature

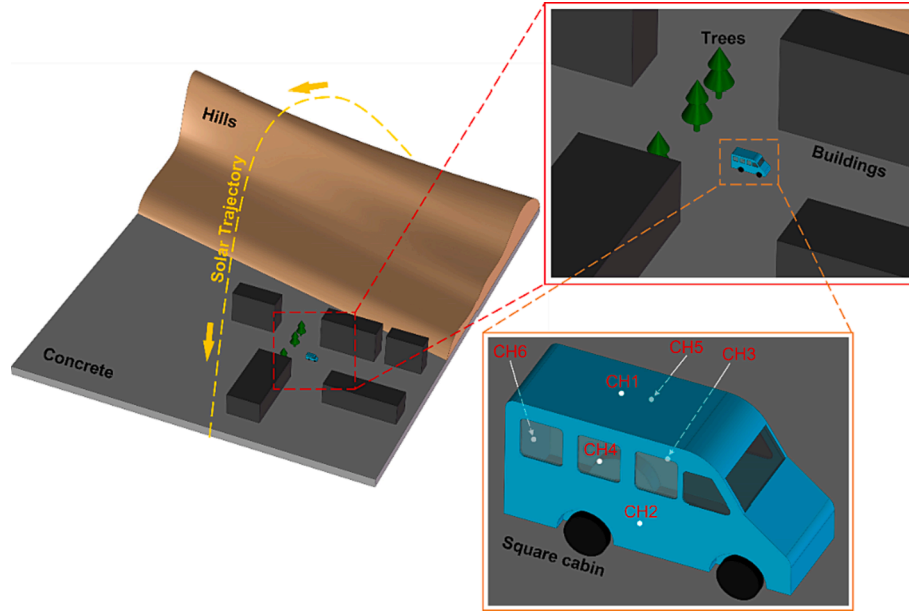


Fig. 2. Simulation model of ground target and its background environment.

fields. The mass, momentum, energy and radiative heat transfer conservation equations used in the calculations are as follows [18]:

$$\nabla \cdot (\rho \vec{V}) = 0 \quad (1)$$

$$\nabla \cdot (\rho \vec{V} Y_j) = -\nabla \cdot \vec{J}_j \quad (2)$$

$$\frac{dI(\vec{r}, \vec{s})}{ds} + aI(\vec{r}, \vec{s}) = a \frac{\sigma T^4}{\pi} \quad (3)$$

where ρ is the density of the gas, $\vec{V}$ is the velocity vector. And the relationship between the radiation intensity I and the heat flux q_r is given by the following equation:

$$\int \frac{dI(\vec{r}, \vec{s})}{ds} d\Omega = \nabla \cdot q_r \quad (4)$$

All terms in the above equations are discretized by the second-order windward format. The realizable $k-\varepsilon$ turbulence model is chosen for the turbulence control equations with the following transport equations and turbulent dissipation rate equations [19].

$$\rho \frac{\partial k}{\partial t} + \rho u \cdot \nabla k = \nabla \cdot \left(\left(\mu + \frac{\mu_T}{\sigma_k} \right) \nabla k \right) + P_k - \rho \varepsilon \quad (5)$$

$$\rho \frac{\partial \varepsilon}{\partial t} + \rho u \cdot \nabla \varepsilon = \nabla \cdot \left(\left(\mu + \frac{\mu_T}{\sigma_\varepsilon} \right) \nabla \varepsilon \right) + C_1 \rho S \varepsilon - C_2 \rho \frac{\varepsilon^2}{k + \sqrt{\nu \varepsilon}} \quad (6)$$

where k is the turbulence kinetic energy, ε is the dissipation rate, the value of the constants are $\sigma_k = 1.0, \sigma_\varepsilon = 1.2, C_2 = 1.9$, and $C_1 = \max\{0.43, \frac{\eta}{5+\eta}\}$, $\eta = (2E_{ij}E_{ij})^{1/2}k/\varepsilon$, $E_{ij} = \frac{1}{2}(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i})$, $\mu_t = \rho C_\mu \frac{k^2}{\varepsilon}$, $C_\mu = \frac{1}{A_0 + A_S U^* k/\varepsilon}$, $A_0 = 4.0$, $A_S = \sqrt{6} \cos \varphi$, $\varphi = \frac{1}{3} \cos^{-1}(\sqrt{6} W)$, $W = \frac{E_{ij}E_{jk}E_{kl}}{(E_{ij}E_{ij})^{1/2}}$, $U^* = \sqrt{E_{ij}E_{ij} + \tilde{\Omega}_{ij}\tilde{\Omega}_{ij}}$, $\tilde{\Omega}_{ij} = \Omega_{ij} - 2\varepsilon_{ij}k\omega_k$, $\Omega_{ij} = \tilde{\Omega}_{ij} - \varepsilon_{ij}k\omega_k$, where Ω_{ij} represents the rotation effect and is the tensor of average rotation rate observed in the reference frame with angular velocity ω_k .

Mesh division

The ground target and the environmental model are relatively variable in scale, thus different meshing strategies are used for different

regions. The global use of unstructured mesh for the solid region is shown in Fig. 3(a), and the mesh refinement is performed on the surface of the ground target and the surrounding area to obtain more detailed temperature distribution results. The ground surface is a two-layer swept mesh, and the total number of meshes in the solid region is 330,000 with an average mesh mass of 0.7. The main ground target studied in this paper is the van, and the flow field calculation of the whole environment requires a lot of resources and time, so the fluid model is simplified to set up a fluid region around the van only, with a length of 3 times the length of the van, a width of 4 times the width of the van, and a height of 3 times the height of the van.

The fluid region mesh is shown in Fig. 3(b), and the fluid region is divided into near-body and far-body regions, where the near-body uses an unstructured mesh, and the mesh refinement is also used in the region near the van surface. The far body region uses a structured hexahedral mesh, which can significantly optimize the computation time. In addition, the fluid is set up with seven boundary layers on the van surface and ground contact surface with a stretching factor of 1.2. The total number of meshes in the fluid region is 370,000, and the average mesh mass is 0.72.

Boundary conditions for different seasons

In order to study the thermodynamic and infrared radiation characteristics of the ground target in different seasons, different boundary conditions are set to simulate the four seasons. Parameters such as ambient temperature, ambient wind speed, solar altitude angle, and sunshine duration are prescribed to vary to define the environmental variation in different seasons. Since the environmental parameters are similar in spring and autumn, only spring is calculated. The geographical location for the simulation calculation was set to 118.84°E, 32.03°N, and the time zone +8. The spring, summer, and winter times were chosen on April 19, July 19, and February 5, 2021, respectively. According to the mesh division result, the side of the hill is set as the sun rising direction and the right side of the van is set as the sun falling direction. The surface in front of the van in the fluid domain is the wind field incoming flow surface, and the other surfaces are set as open boundaries. In addition, the bottom of the ground is set to the constant temperature $T_{average}$, which is usually the average temperature of the day.

In specific simulation calculations, it is often difficult to know the

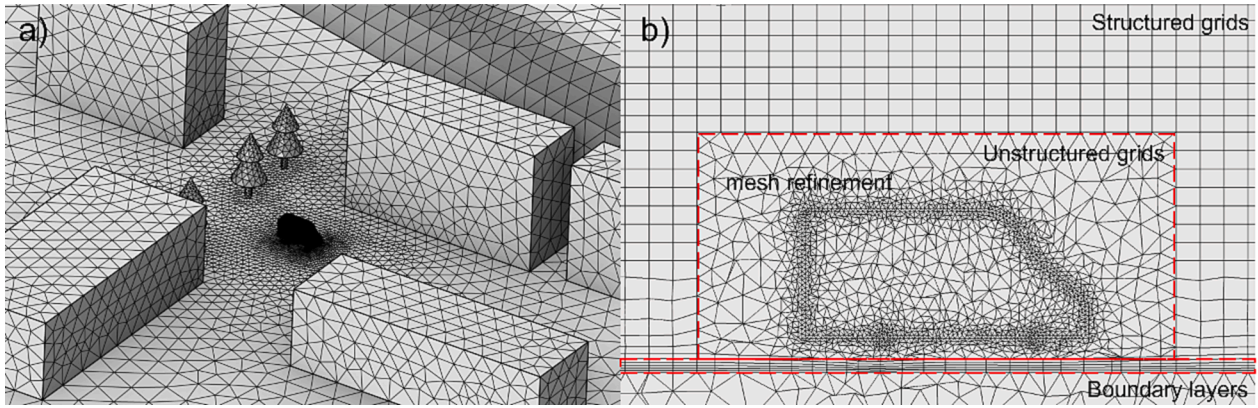


Fig. 3. Mesh of ground target and its background environment: a) Solid domain mesh; b) Fluid domain mesh.

atmospheric temperature and ambient wind speed at a certain time, but predictions can be made from meteorological information such as local average temperature, extreme temperature difference, maximum and

minimum wind speed. Atmospheric temperature usually has a cyclical pattern from year to year and day to day. In a given day, the temperature curve with time can be approximated as a simple harmonic wave, and

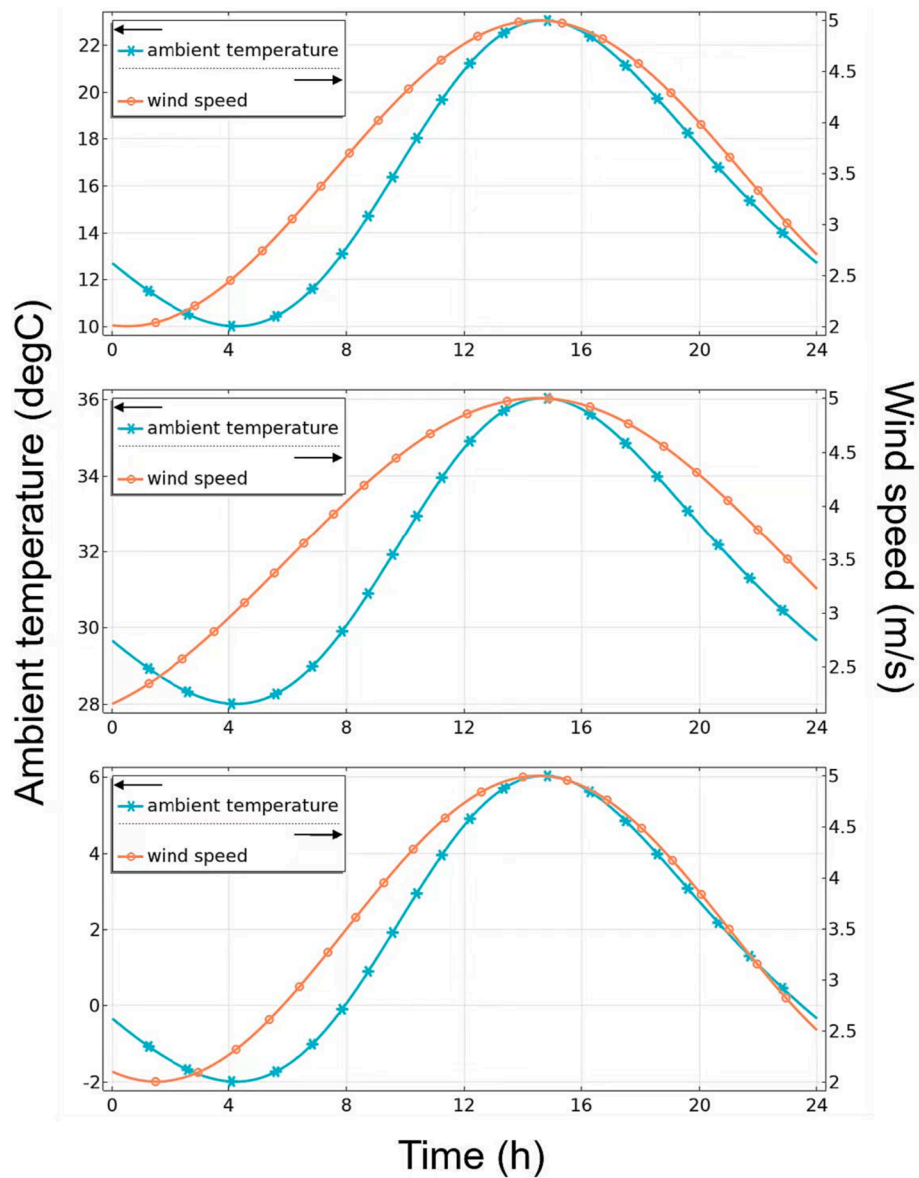


Fig. 4. Ambient temperature and wind speed in different seasons: a) Spring; b) Summer; c) Winter.

using the second-order Fourier series expansion, the temperature at time $t(h)$ of the day can be expressed as [20]:

$$T_a(t) = T_{average} + 0.491 \times \Delta T \times \cos\left[\frac{\pi}{12} \times (t - 15.32)\right] + 0.058 \times \Delta T \times \cos\left[\frac{\pi}{6} \times (t - 1.22)\right] \quad (7)$$

where $T_{average}$ is the approximate average temperature of the day:

$$T_{average} = 0.478 \times T_{amax} + 0.522 \times T_{amin} \quad (8)$$

ΔT is the extreme temperature difference:

$$\Delta T = T_{amax} - T_{amin} \quad (9)$$

The other coefficients in the equation are determined from the information in the Chinese meteorological data network combined with the simulation locations using the least squares method.

The wind speed has a large effect on the convective heat transfer coefficient between the target surface and the air. According to Ephrath et al, the wind speed at moment t is calculated as [20]:

$$v(t) = v_{min} + v_{max} \times \sin\left(2 \times \pi \times \frac{t - t\omega_{1,2}}{SF_{1,2}}\right) \quad (10)$$

$$t\omega_1 = T_1; t\omega_2 = 2 \times (T_2 - T_3); SF_1 = 4 \times (T_2 - T_1); SF_2 = 4 \times (T_3 - T_2) \quad (11)$$

where the value of T_1 , T_2 , T_3 is related to the geographical location and meteorological conditions. T_1 is approximately 1 h after sunrise, T_2 is 14.5 h, and T_3 is 1 h after sunset according to the simulation coordinates and the information from the Chinese meteorological data network. The ambient temperature and ambient wind speed curves for different seasons are shown in Fig. 4. It can be seen that the maximum daily temperature in spring is 23 °C around 15 h and the minimum daily temperature is 10 °C around 4.5 h in the morning. The maximum daily temperature in summer is 36 °C and the minimum temperature is 28 °C. And the daily maximum temperature in winter is 6 °C and the minimum temperature is -2 °C.

Verification

The coupled simulation method is validated using a standard CUBI model with the mesh division shown in Fig. 5. The CUBI model is made of low carbon steel (4 mm thick plate). The total height and length of CUBI is 1 m and its width is 0.5 m. The back side of the steel plate (model hollow with insulation material on the inner face) is insulated with 10 mm thick foam polyurethane. The density of carbon steel material is 7768.98 (kg/m³), specific heat capacity is 460.967 (J/kg·K), and thermal conductivity is 52.019 (W/m·K). The second layer is a thermal insulator (10 mm foam polyurethane) with a density of 20 (kg/m³), specific heat capacity of 1350 (J/kg·K) and thermal conductivity of 0.035 (W/m·K). The solar absorptance $\alpha = 0.85$ and tropical emissivity $\varepsilon = 0.93$ for the outer surface. The emissivity of the inner surface of the polyurethane is 0.9. The simulation time is May 17 - May 18, 2005. The environmental parameters on the day of the case validation simulation are shown in Fig. 6 [21].

Fig. 7 and Fig. 8 show the comparison between the calculated data and the experimental data at two points on the CUBI model, the orange line is the experimental data while the blue line is the simulation data, and it can be found that the temperature values and change trends of both are basically consistent. Since the experiment only measured the solar irradiance of the day and did not distinguish between direct solar radiation and solar scattering, it was not possible to obtain the direct solar irradiance and scattered irradiance at noon of the day, so the radiation from the sun could only be set to all direct solar radiation without considering solar scattering according to its measured environmental data. Therefore, the experimental data and the simulation

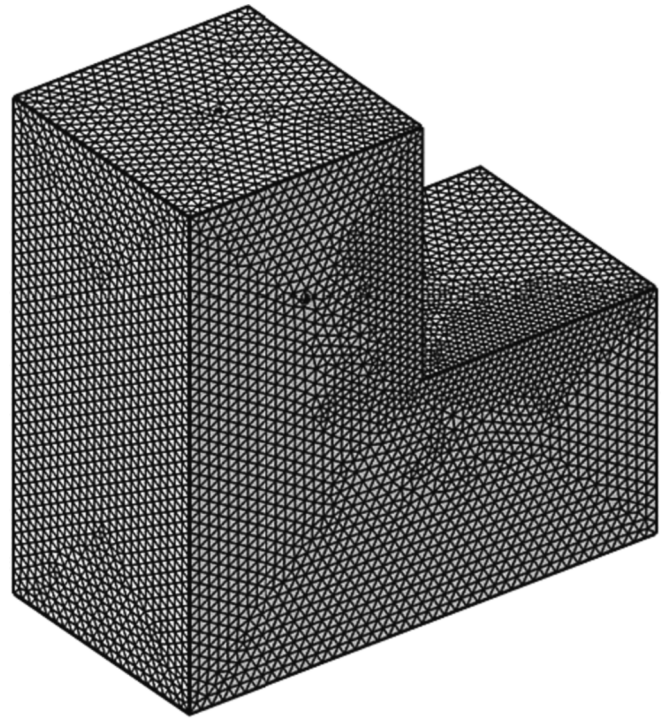


Fig. 5. CUBI model and its mesh.

results produce some deviation, making the simulation data of the CUBI model higher in temperature at sunrise in the morning and sunset in the evening.

Results and discussions

Thermodynamic images of ground target and the surrounding environment

The thermodynamic distribution characteristics of the ground target and the surrounding environment were investigated in spring, summer and winter using the coupled multi-physics field simulation method. 6 time points 4:00, 8:00, 12:00, 16:00, 20:00 and 24:00 were selected to observe the temperature change patterns of the ground target and the surrounding environment based on the calculation results of the full time period 24 h.

Fig. 9 shows an image of the temperature distribution of the ground target in spring, and the maximum displayed temperature is set to 40 °C and the minimum is set to 10 °C for ease of observation. At 0–8 h, the sun rises from the left direction of the van, raising the temperature of its east-facing face, while leaving a shadow with lower temperature on the ground. At 12 h the top of the van is exposed to direct sunlight and the temperature approaches a maximum of 40 °C. At 16 h, both the top of the van and the area on the right face except for the windows reached about 40 °C, and the temperature of the surrounding buildings and trees was slightly lower than that of the van. At 20–24 h, the whole ambient temperature decreased and gradually turned back to purple. At 24 h, the ground temperature distribution is still uneven, with lower temperature in places that produce shadows for longer periods of time during the day.

Fig. 10 shows an image of the temperature distribution of the ground target in summer, with the highest displayed temperature set to 50 °C and the lowest set to 28 °C in the legend. At 8 h, the temperature of the rear part of the van is higher than the front part of the van because of the building shade. Compared to 12–16 h in spring, the whole ambient temperature is higher due to the increased intensity of solar radiation, and the boundary between the area exposed to direct solar radiation and the shaded area is clearly defined. At 20–24 h, it can be seen that the tire temperature of the van drops faster compared to the van body

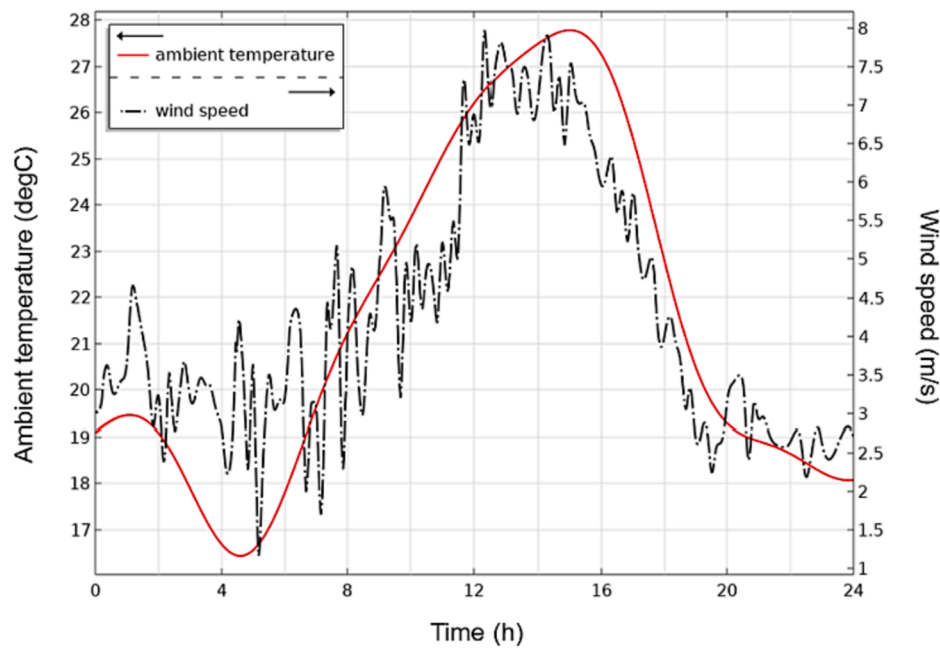


Fig. 6. Ambient temperature and wind speed for verification.

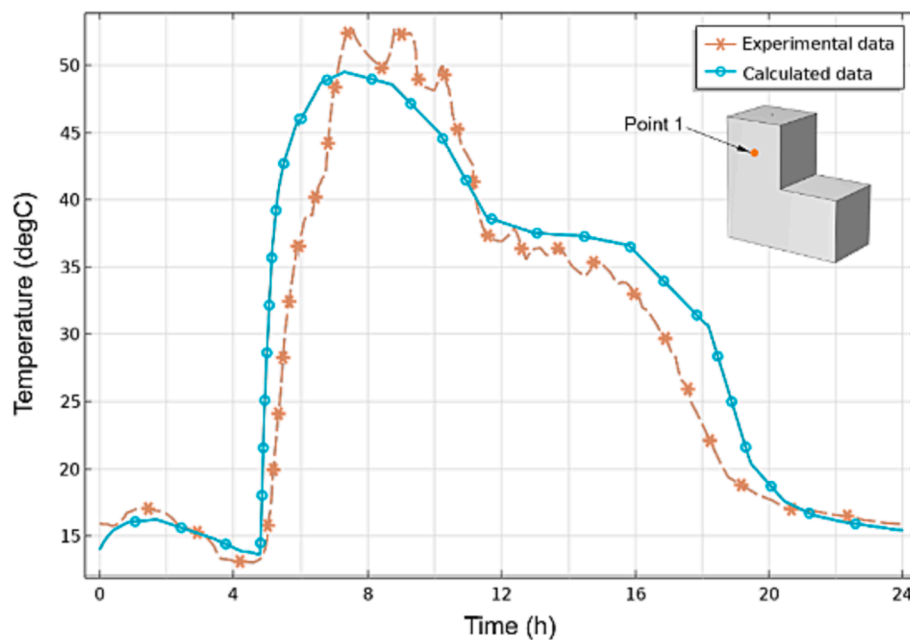


Fig. 7. Experimental measured temperature and simulated temperature at point 1 on the CUBI model.

temperature, which is mainly due to the different material physical properties.

Fig. 11 shows an image of the temperature distribution of the ground target in winter, with the highest displayed temperature set to 20 °C and the lowest set to −5 °C. At 4–8 h, the surface of the van is slowly warmed by atmospheric scattered radiation due to the very low solar altitude angle and complete shading by buildings. At 12 h, shadows of different temperatures can be observed on the ground. At 16 h, the temperature of the south-facing wall of the buildings can be seen to be higher than the other faces. At 20–24 h, the overall ambient temperature drops rapidly, with the lowest temperature dropping below zero degrees.

Thermodynamic images of the van

In order to further study the thermodynamic characteristics of the ground target under different seasons, the temperature distribution changes of the van under different time periods were analyzed. Fig. 12 shows the image of the temperature distribution of the van in spring, and the maximum temperature of the legend is set to 45 °C and the minimum is set to 10 °C for the convenience of observation. At 8 h, the area below the window on the right surface of the van produces shallow light spots, which is due to the translucent nature of the glass, which allows some of the wavelengths in the solar radiation to pass directly through the glass and thus irradiate onto the inner wall of the van, thus increasing the temperature of the irradiated area and forming square light spots. In

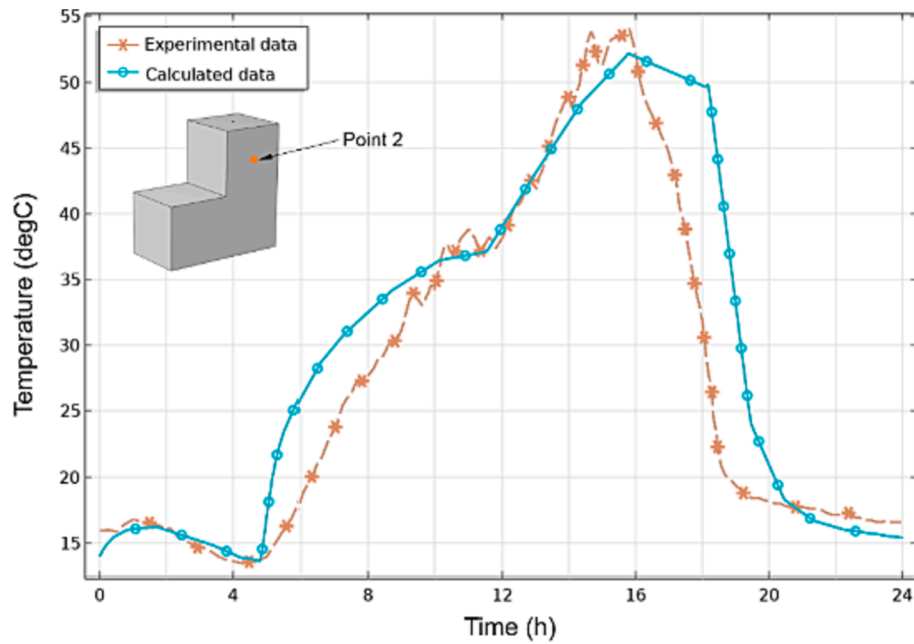


Fig. 8. Experimental measured temperature and simulated temperature at point 2 on the CUBI model.

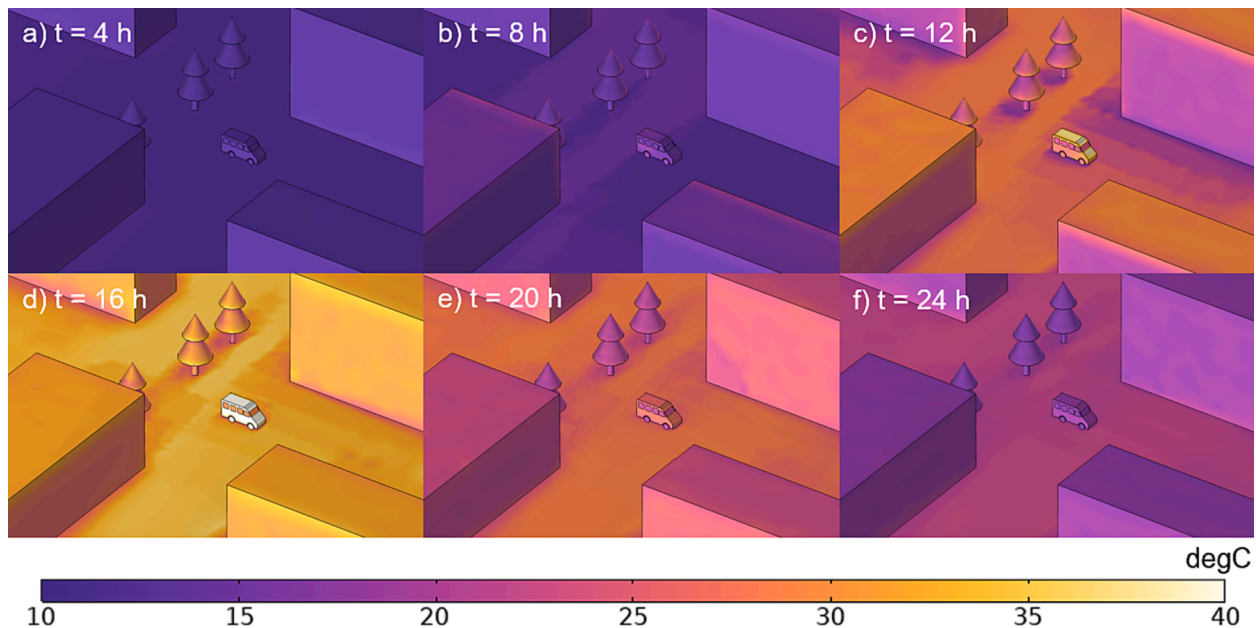


Fig. 9. Temperature distribution of the ground target and its background at different moments in spring: a) $t = 4$ h; b) $t = 8$ h; c) $t = 12$ h; d) $t = 16$ h; e) $t = 20$ h; f) $t = 24$ h.

addition, the temperature of the top and left side surfaces is higher than the right side and the obscured part of the front of the van. At 12 h, the temperature of the top surface of the van increased to about 40 °C and was higher in the middle and lower around. At 16 h, the temperature of the right wheel reached 45 °C, the window was about 27 °C, and the other areas were 35–40 °C. At 20 h, the temperature on the right side of the wall of the van was higher than the top side.

Fig. 13 shows an image of the temperature distribution of the van in summer, with the maximum temperature set to 50 °C and the minimum set to 28 °C in the legend. At 4 h, the temperature of the van is consistent with the surrounding ground temperature. At 8 h, three distinct glass-shaped square light spots can be seen on the right surface of the van. At 12 h, the top of the square cabin can be seen to be subjected to strong

direct solar radiation, with the temperature reaching 50 °C and white. At this time, the ground shadow has a clear layering effect, the closer the area to the van the lower the temperature and the darker the color. At 16 h, both the top and the right side of the van are exposed to solar radiation and the overall temperature is higher, reaching 50 °C except for the windows. At the same time, the shadows on the ground on the right side of the van become lighter, and new shadows are produced on the ground on the left side.

Fig. 14 shows an image of the temperature distribution of the van in winter, with the maximum temperature set to 20 °C and the minimum set to −5 °C in the legend. At 4–8 h, the van is shaded by the building next to it and not exposed to direct solar radiation, and the temperature change is not obvious. At 12 h, the temperature at the top of the van

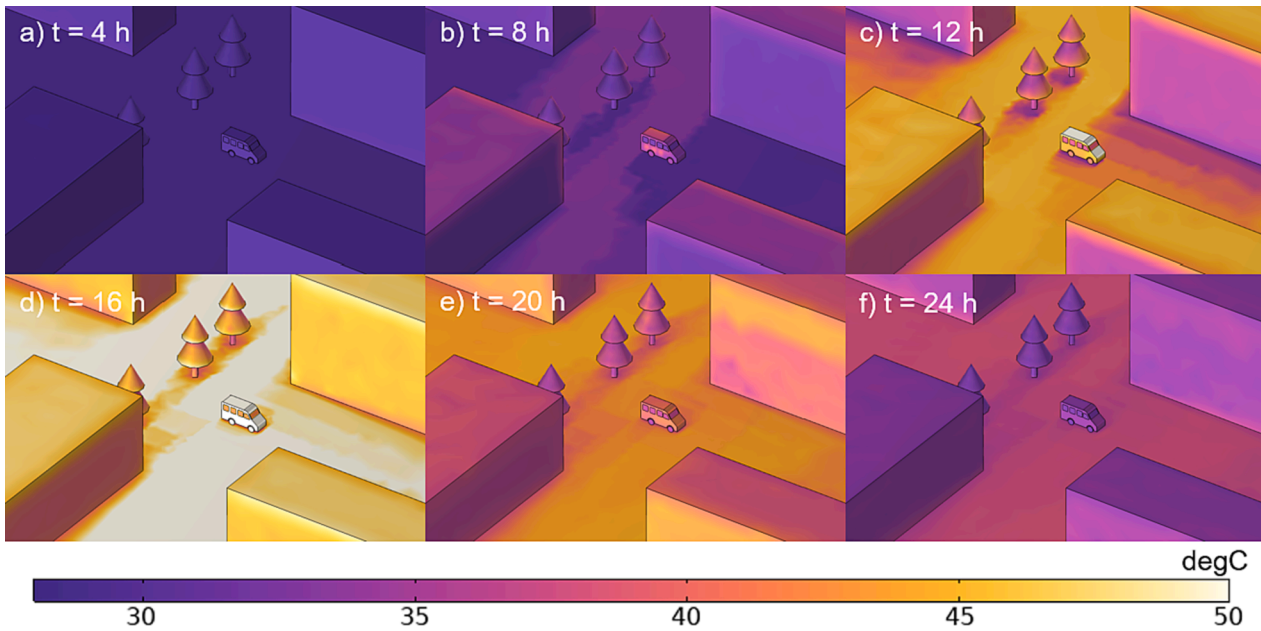


Fig. 10. Temperature distribution of the ground target and its background at different moments in summer: a) $t = 4$ h; b) $t = 8$ h; c) $t = 12$ h; d) $t = 16$ h; e) $t = 20$ h; f) $t = 24$ h.

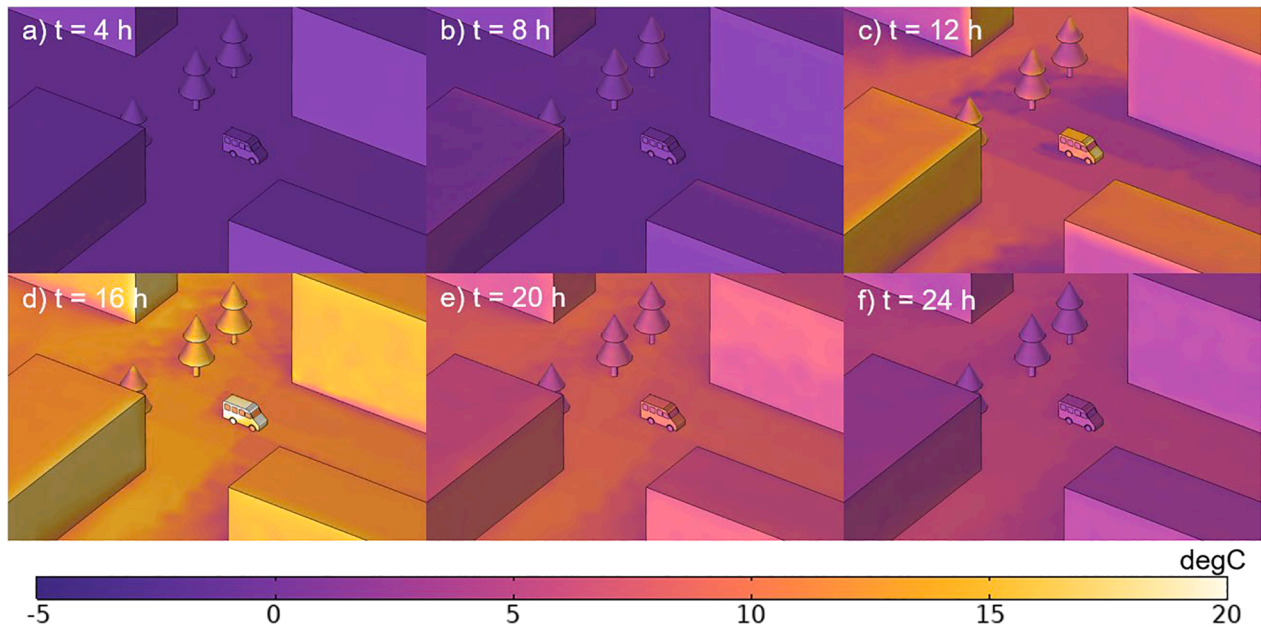


Fig. 11. Temperature distribution of the ground target and its background at different moments in winter: a) $t = 4$ h; b) $t = 8$ h; c) $t = 12$ h; d) $t = 16$ h; e) $t = 20$ h; f) $t = 24$ h.

increases, while a rectangular light spot appears on the right surface, which is formed by the direct sunlight passing through the front glass and finally reaching the inner wall surface on the right side of the van. It can also be observed that the direction of the shadow formed by the van on the ground at this time is from south to north. At 16 h, there is a building blocking the head area of the van, which makes the temperature in the rear half of the van higher than the head area and forms a clear dividing line on the right surface. At 20–24 h, the van temperature decreases rapidly, but is still higher than the ground temperature.

Comparison of temperature changes at monitoring points

The temperature data of six monitoring points on the surface of the

van are recorded in different seasons, and Fig. 15 shows the comparison of temperature changes of these six points in spring, summer and winter for 24 h a day, respectively. The orange line indicates the simulation data in summer, the blue line indicates the simulation data in spring, and the purple line indicates the simulation data in winter. The locations of the monitoring points are shown in Fig. 2, where Fig. 15(a)-CH1 is located on the top side of the van, Fig. 15(b)-CH2 is located on the right side of the van, Fig. 15(c)-CH3 is located on the left side of the van, Fig. 15(d)-CH4 is located on the right side of the van window, Fig. 15(e)-CH5 is located on the left side of the van window, and Fig. 15(f)-CH6 is located at the back of the van.

By observing the temperature curves of different seasons shown in Fig. 15, it can be concluded that the temperature of the ground target is

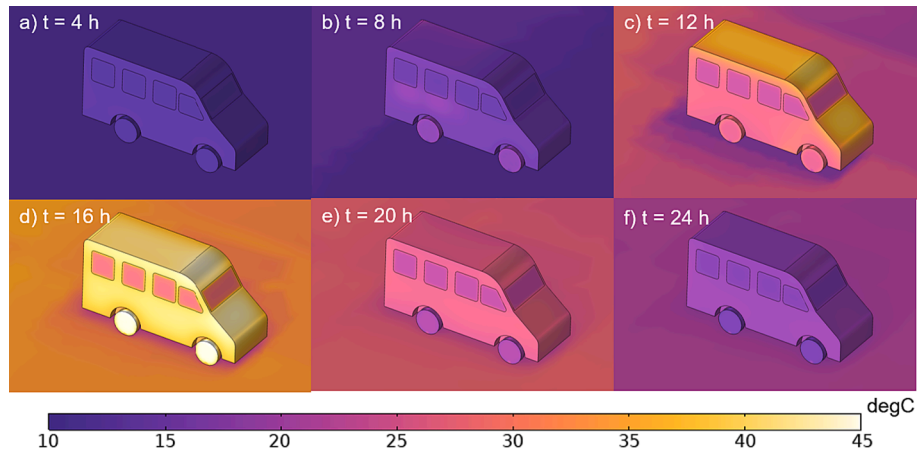


Fig. 12. Temperature distribution of the van at different moments in spring: a) $t = 4$ h; b) $t = 8$ h; c) $t = 12$ h; d) $t = 16$ h; e) $t = 20$ h; f) $t = 24$ h.

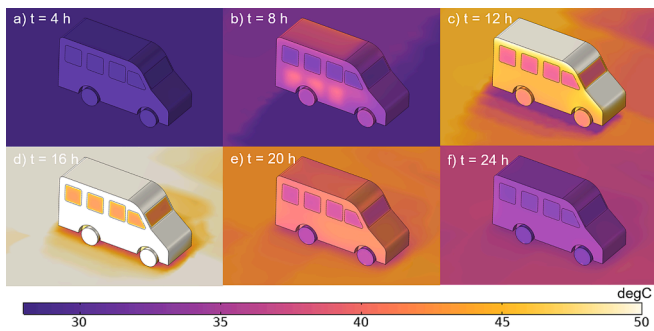


Fig. 13. Temperature distribution of the van at different moments in summer: a) $t = 4$ h; b) $t = 8$ h; c) $t = 12$ h; d) $t = 16$ h; e) $t = 20$ h; f) $t = 24$ h.

closely related to the ambient temperature. The ambient temperature is high in summer, and therefore the surface temperature of the van is also the highest. And the ambient temperature is low in winter, and the surface temperature of the van is also the lowest. In addition, when there is no shading from surrounding buildings, direct sunlight and altitude angle changes directly affect the trend of ground target temperature. When the seasons are the same and the locations are different, the east monitoring point warms up earlier than the west monitoring point; when the seasons are different and the locations are the same, the van

warms up earliest in summer due to the earlier sunrise time and the van warms up latest in winter due to the latest sunrise time.

According to Fig. 15(a), the top surface temperature of the van first increases and then decreases, reaching a peak at about 15 h. The daily maximum temperature is 57 °C in summer, 43 °C in spring, and only 20 °C in winter. The daily minimum temperature occurs around 5 h, where 28 °C in summer, 11 °C in spring, and −1 °C in winter.

Fig. 15(b) shows the temperature profile of the right point of the van. Because it faces south, the peak temperature appears later, at around 17 h. In addition, the inflection point that occurs at 16 h in winter is due to the shading by the surrounding buildings.

Fig. 15(c) shows the temperature profile of the left point of the van, and it can be seen that in summer and spring the temperature starts to rise at around 5 h and reaches the peak temperature at 16 h. In winter, the highest temperature occurs at 15 h.

Fig. 15(d) and (e) show the temperature change curves of the monitored points on the left and right windows of the van, respectively, which are slightly slower compared to the other points. The peak temperature is slightly higher than 40 °C in summer, about 27 °C in spring, and slightly lower than 10 °C in winter.

Fig. 15(f) shows the temperature profile at the rear side point of the van, which has a similar trend to Fig. 15(a), but rises earlier and cools more slowly. And other data are summarized in Table 2.

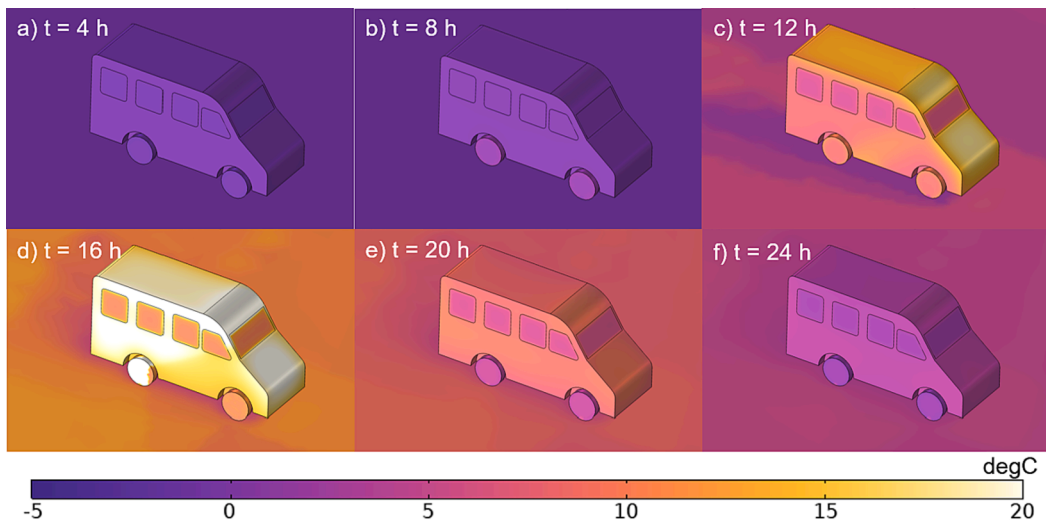


Fig. 14. Temperature distribution of the van at different moments in winter: a) $t = 4$ h; b) $t = 8$ h; c) $t = 12$ h; d) $t = 16$ h; e) $t = 20$ h; f) $t = 24$ h.

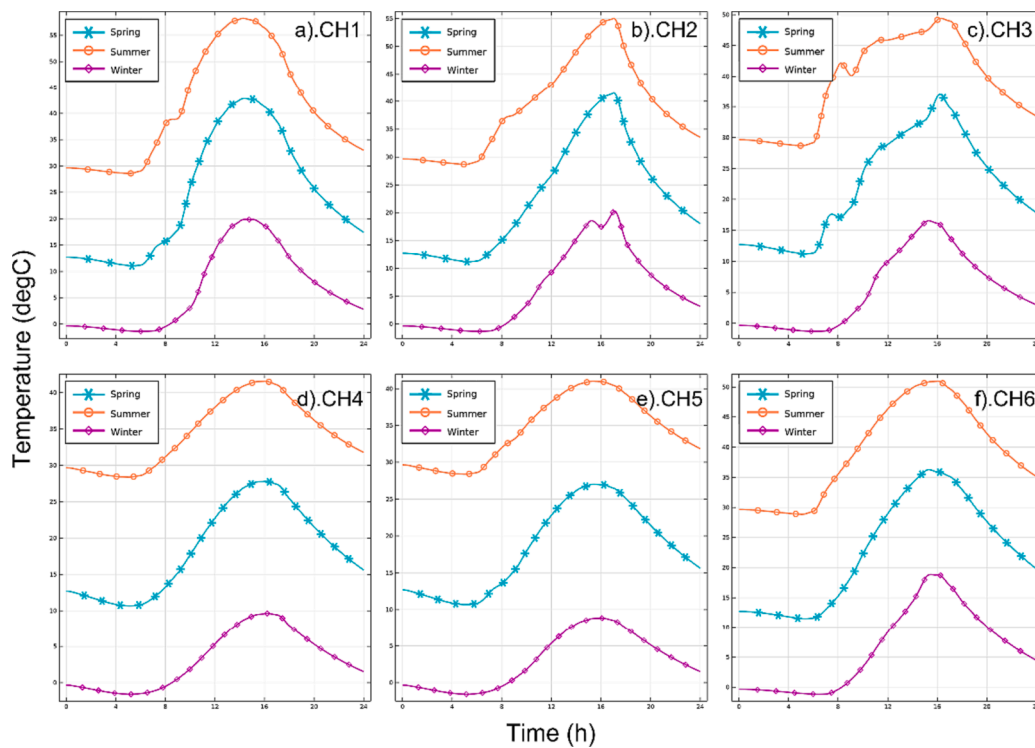


Fig. 15. Temperature change curves of different monitoring points on the van: a) CH1; b) CH2; c) CH3; d) CH4; e) CH5; f) CH6.

Table 2
Temperature distribution of different monitoring points.

	Tmin(Sun/Spr/Win)	Tmax(Sum/Spr/Win)	Diff(Sum/Spr/Win)
CH1	29/11/-1(°C)	58/42.5/20(°C)	29/31.5/21(°C)
CH2	29/11/-1(°C)	55/41/20(°C)	26/30/21(°C)
CH3	29/11/-1(°C)	50/37/16(°C)	21/26/17(°C)
CH4	29/12/-1(°C)	51/36/19(°C)	22/24/20(°C)
CH5	28/11/-2(°C)	42/27.5/9.5(°C)	14/16.5/11.5(°C)
CH6	28/11/-2(°C)	42/27/8(°C)	14/16/10(°C)

Conclusions

Based on thermodynamics and computational fluid dynamics, this study simulates the thermodynamic properties of ground target and its background environment under different seasons using coupled multi-physics fields. By observing and comparing the temperature distribution images and the temperature profiles of the monitoring points, the following conclusions are summarized.

1. Both the ground target and the background exhibit different thermodynamic characteristics under different seasons, with the van temperature higher than the concrete ground during daytime and lower than the ground at night.
2. The temperature distribution on the ground target with different structures and different materials differed significantly.
3. The temperature of the ground target is closely related to the change of seasons, where the maximum temperature reaches 57 °C in summer and only 20 °C in winter.
4. Due to the larger solar altitude angle in summer, the ground target warms up earlier in summer than in spring and winter.

CRediT authorship contribution statement

Zhaobing Jiang: Conceptualization, Resources, Writing – original draft, Supervision, Funding acquisition. **Bing Wang:** Writing – review & editing, Visualization. **Zhiyuan Ren:** Data curation, Visualization,

Methodology, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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